Transcriptional Noise and Rare Phenotypic Switching in Epithelial–Mesenchymal Heterogeneity

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Organised by

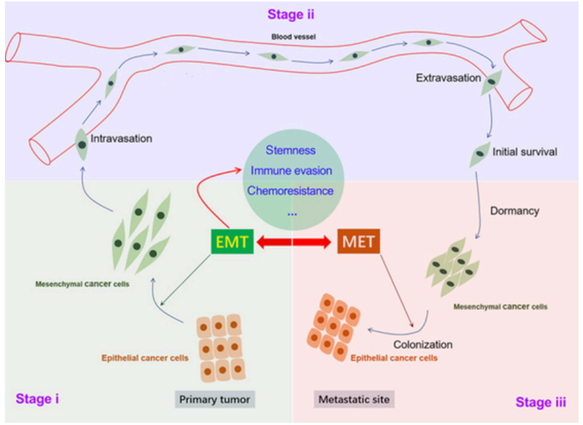
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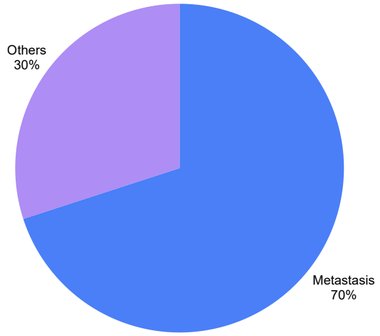
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BACKGROUND

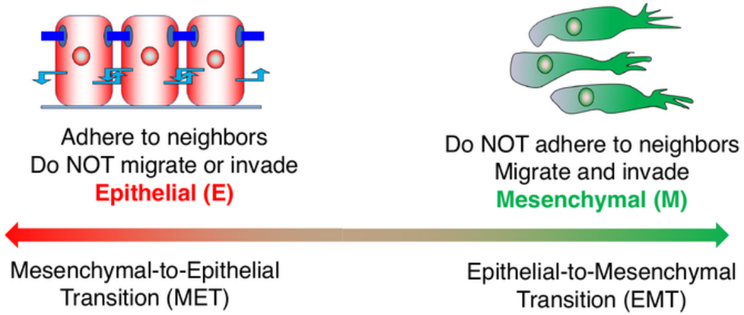
Why metastasis needs to be addressed?



Cells can separate from the primary organ, enter the circulatory and lymphatic system, and invade other organs

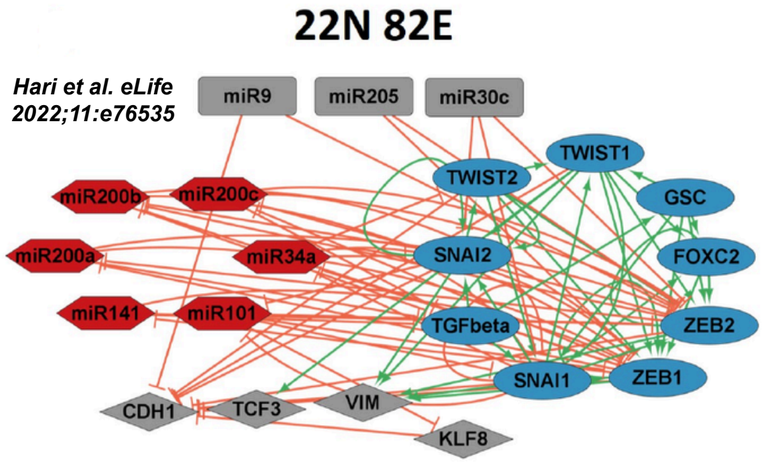


Metastasis accounts for 70-90% of solid tumor related deaths



Metastasized tumor greatly differ from primary tumor in terms of phenotype even though they originate from the same genotype

What do we currently know?



EMP network with 22 Nodes & 82 Edges

- TEAMS network regulates epithelial mesenchymal plasticity
- Boolean simulations suggest that network is robust to external perturbations
- Terminal states are stable and hybrid states are vulnerable to perturbations

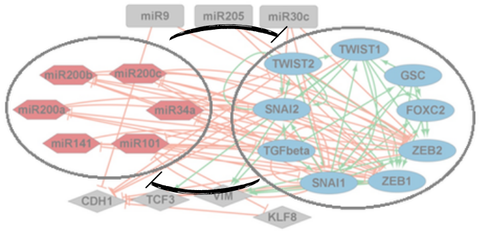
What's the unresolved problem?

- Cancer cells exhibit epithelial-mesenchymal heterogeneity
- There must be SOMETHING that breaks the robustness of the network and makes it vulnerable to perturbations
- How can cells spontaneously switch between terminal & hybrid phenotypic states despite **robustness** of the underlying network?

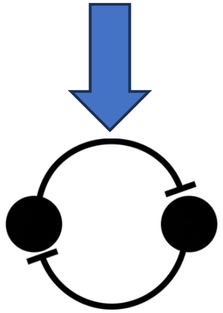
HYPOTHESIS

- **Network size and connectivity influence the emergence of heterogeneity.**
- **Transcriptional noise may contribute to spontaneous switching**

Computational Framework & Methodology



Simplified



production degradation

$$\frac{dm_1(t)}{dt} = \{g\cdot(1-(1-\lambda)\cdot H(m_2)) + \sigma_{birth,1}\cdot N(0,1)\} - \{k\cdot m_1 + \sigma_{death,1}\cdot N(0,1)\}$$

$$\frac{dm_2(t)}{dt} = \{g\cdot(1-(1-\lambda)\cdot H(m_1)) + \sigma_{birth,2}\cdot N(0,1)\} - \{k\cdot m_2 + \sigma_{death,2}\cdot N(0,1)\}$$

$$\sigma_{birth,i} = \sqrt{g\cdot(1-(1-\lambda)\cdot H(m_j))}$$

$$\sigma_{death,i} = \sqrt{k\cdot m_i}$$

$$H(m) = \frac{m^n}{m^n + K^n}$$

Toggle switch → **Chemical Langevin Equation** → **Parameters generated using RACIPE** → **Simulation & Statistical Analysis**

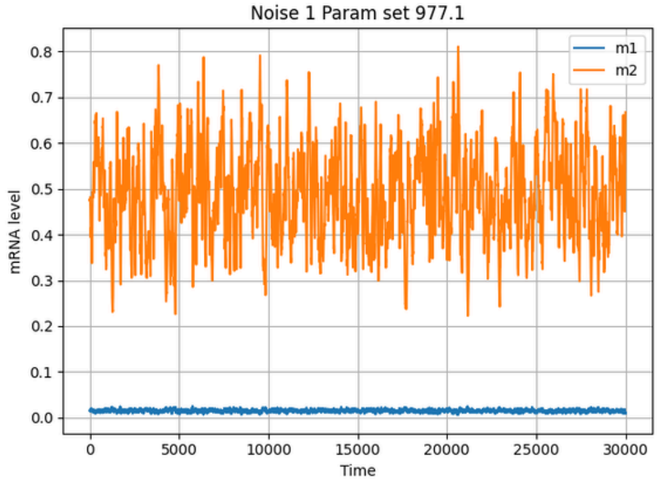
- Toggle switch is a simplified representation of the “teams” network
- Regulation → Shifted hill function
- Gaussian noise assumption
- Std. dev. is a function of copy number

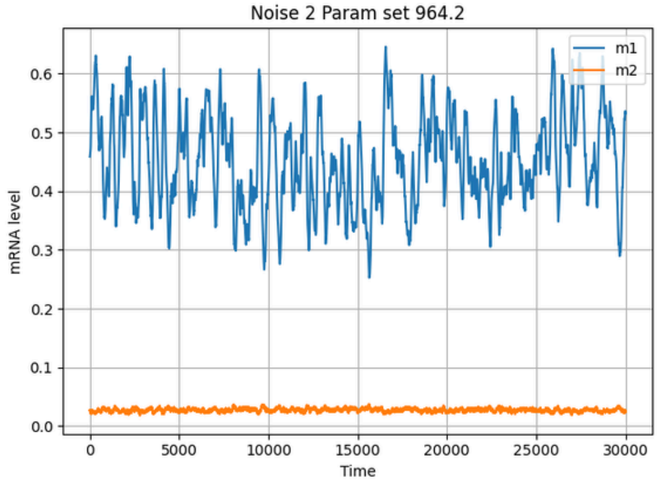
- Threshold (λ) → using half-functional rule
- Only bistable parameters considered
- ODE simulation to obtain steady states

- Noise in one nodes vs both nodes
- 30,000 time points
- Calculated Jensen Shanon Distance
- Distribution of relative expression

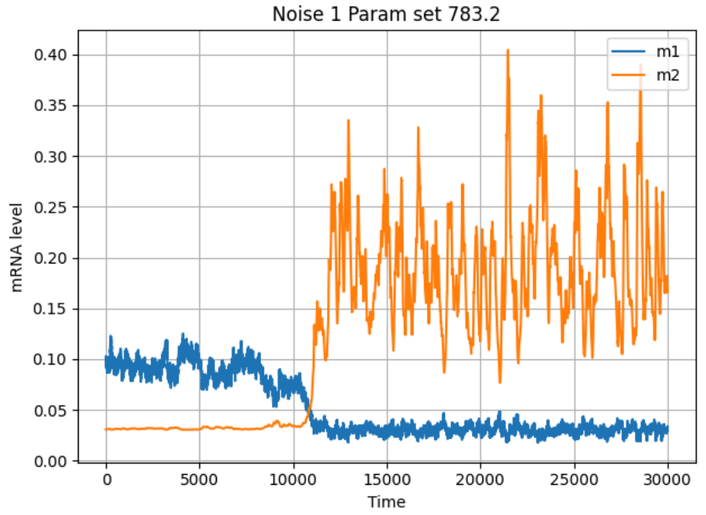
Results I: Expression Dynamics Under Transcriptional Noise

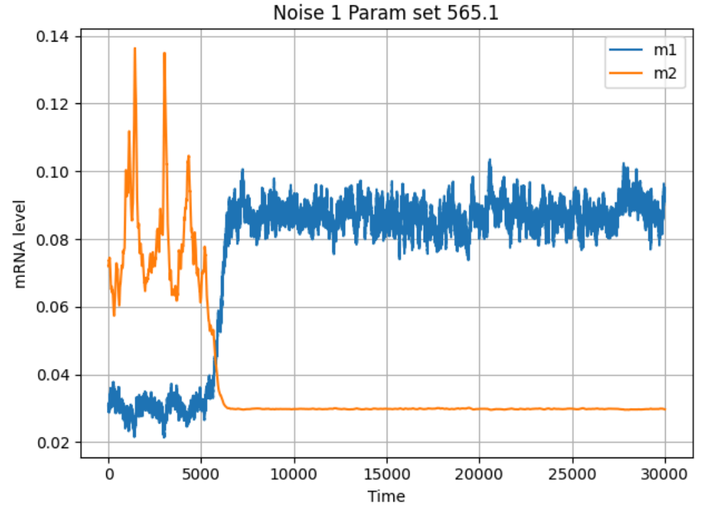
I. Terminal states maintained



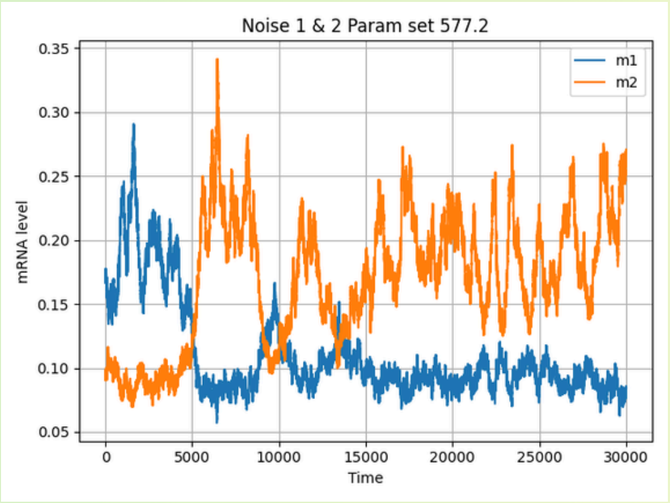


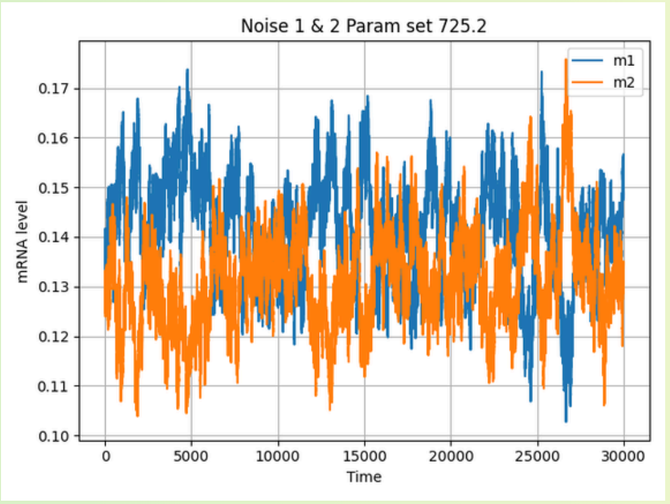
II. Single switching event





III. (RARE) spontaneous switching (Hybrid states) (1%)

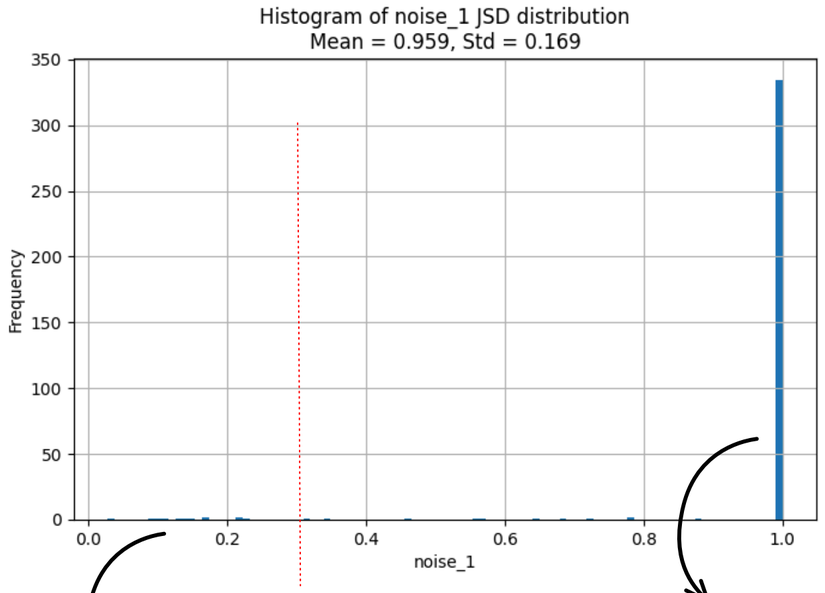




Results II: Statistical Evidence of Hybrid States

Are the expression dynamics dependent on the initial conditions??

High JSD – Distinct distributions (dependent)
Low JSD – Overlap b/w distributions (independent)



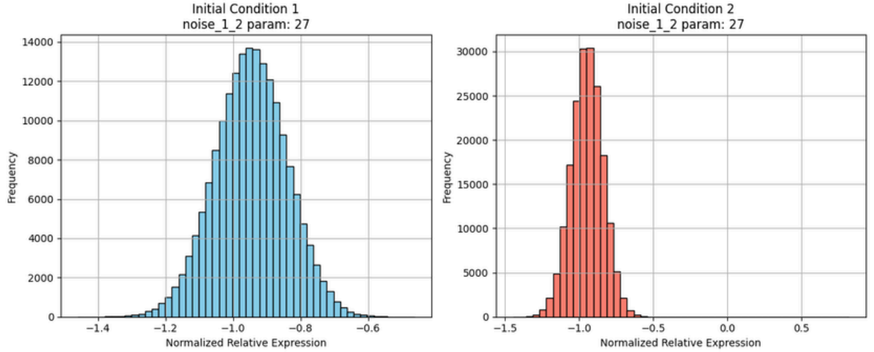
small fraction of parameters lead to overlapping distributions

majority of parameter sets show high JSD

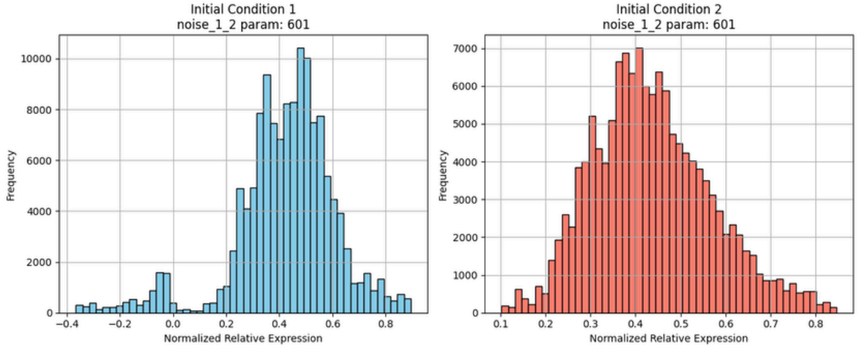
This is not necessarily an evidence of hybrid states, but that hybrid states could be possible under certain regimes

Relative expression analysis (norm. node1 exp – norm. node2 exp)

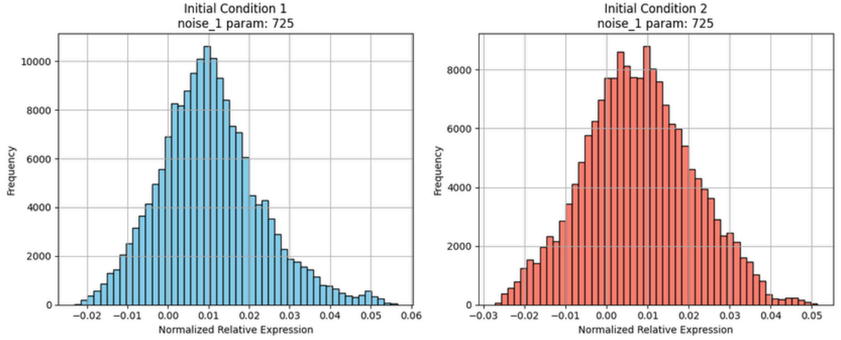
Unimodal distribution
(terminal states maintained)



Bimodal distribution
(switching of terminal states)



Hybrid states
(equal expression
by both nodes)



Conclusions & Future Directions

Key Findings

- Stochastic simulations reveal multiple expression patterns under noise
- Noise-associated switching can occur in simplified toggle-switch.
- Hybrid state expression patterns are very rare and highly dependent on the model parameters

Limitations

- We considered toggle switch – which is a simplified representation of the actual large TEAMS networks
- Gaussian noise is assumed through CLE approximation
- Biological interpretation of these hybrid states requires further validation.

Future Directions

- More **rigorous methods** for statistical analysis
- Extend analysis to **larger** and more **complicated networks**
- Model different noise mechanisms
- Validate predicted switching behaviour against experimental single cell gene-expression data.

Thank You!

Questions ??

Feedback ??